

Find the separation vector $\mathbf{r}$ from the source point $(2, 8, 7)$ to the field point $(4, 6, 8)$. Determine its magnitude (r) , and construct the unit vector $\hat{\mathbf{r}}$.

Solution:

The separation vector $\mathbf{r}$ is the difference between the field point and the source point.

$$\mathbf{r} = (4, 6, 8) - (2, 8, 7) = (2, -2, 1)$$

The magnitude is

$$|\mathbf{r}| = r = \sqrt{2^2 + (-2)^2 + 1^2} = 3$$

and the unit vector is

$$\frac{\mathbf{r}}{r} = \hat{\mathbf{r}} = \frac{1}{3}(2, -2, 1)$$

□
